

Internet of Things - Project 1

Smart Environmental Monitor (Microcontroller Basics)

Objective

To interface a DHT22 temperature and humidity sensor with an ESP32, read environmental parameters every 2 seconds, and display the readings using the Serial Monitor and an I2C LCD while understanding blocking and non-blocking execution.

Implementation

Three implementations were completed:

- Using delay() with Serial Monitor output.
- Using millis() (non-blocking) with Serial Monitor output.
- Using an I2C LCD to display temperature and humidity.

Source code:

Using delay()

```
#include <DHT.h>
#define DHTPIN 4
#define DHTTYPE DHT22

DHT dht(DHTPIN, DHTTYPE);

void setup() {
  Serial.begin(115200);

  dht.begin();

  Serial.println("Smart Environmental Monitor");
}

void loop() {
  float temperature = dht.readTemperature();
  float humidity = dht.readHumidity();

  if (isnan(temperature) || isnan(humidity)) {
```

```

Serial.println("Failed to read from DHT sensor");

} else {

    Serial.print("Temperature: ");
    Serial.print(temperature);
    Serial.print(" °C");

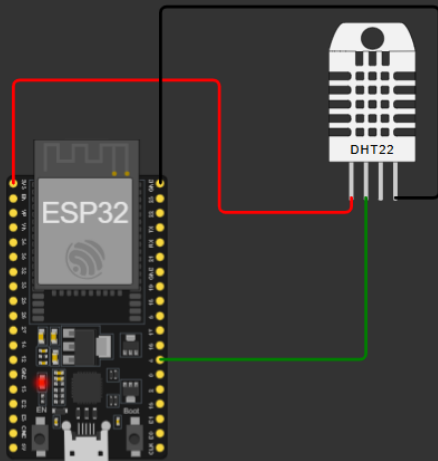
    Serial.print("  Humidity: ");
    Serial.print(humidity);
    Serial.println(" %");
}

delay(2000);
}

```

Circuit Diagram and Output

Simulation
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Smart Environmental Monitor

Temperature: 24.90 °C	Humidity: 31.00 %
Temperature: 24.90 °C	Humidity: 31.00 %
Temperature: 24.90 °C	Humidity: 31.00 %
Temperature: 24.90 °C	Humidity: 31.00 %
Temperature: 24.90 °C	Humidity: 31.00 %
Temperature: 24.90 °C	Humidity: 31.00 %

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Using millis()

```
#include <DHT.h>

#define DHTPIN 4
#define DHTTYPE DHT22

DHT dht(DHTPIN, DHTTYPE);
unsigned long previousMillis = 0;
const unsigned long interval = 2000;

void setup() {
  Serial.begin(115200);

  // Initialize DHT Sensor
  dht.begin();

  Serial.println("-----");
  Serial.println(" Smart Environmental Monitor");
  Serial.println(" DecodeLabs - Project 1");
  Serial.println("-----");
  Serial.println("Reading Temperature & Humidity...");
  Serial.println();
}

void loop() {
  unsigned long currentMillis = millis();

  if (currentMillis - previousMillis >= interval) {

    previousMillis = currentMillis;

    float humidity = dht.readHumidity();
    float temperature = dht.readTemperature();

    if (isnan(humidity) || isnan(temperature)) {

      Serial.println(" Failed to read from DHT sensor!");
      Serial.println();

    } else {

      Serial.println("===== Sensor Readings =====");
```

```

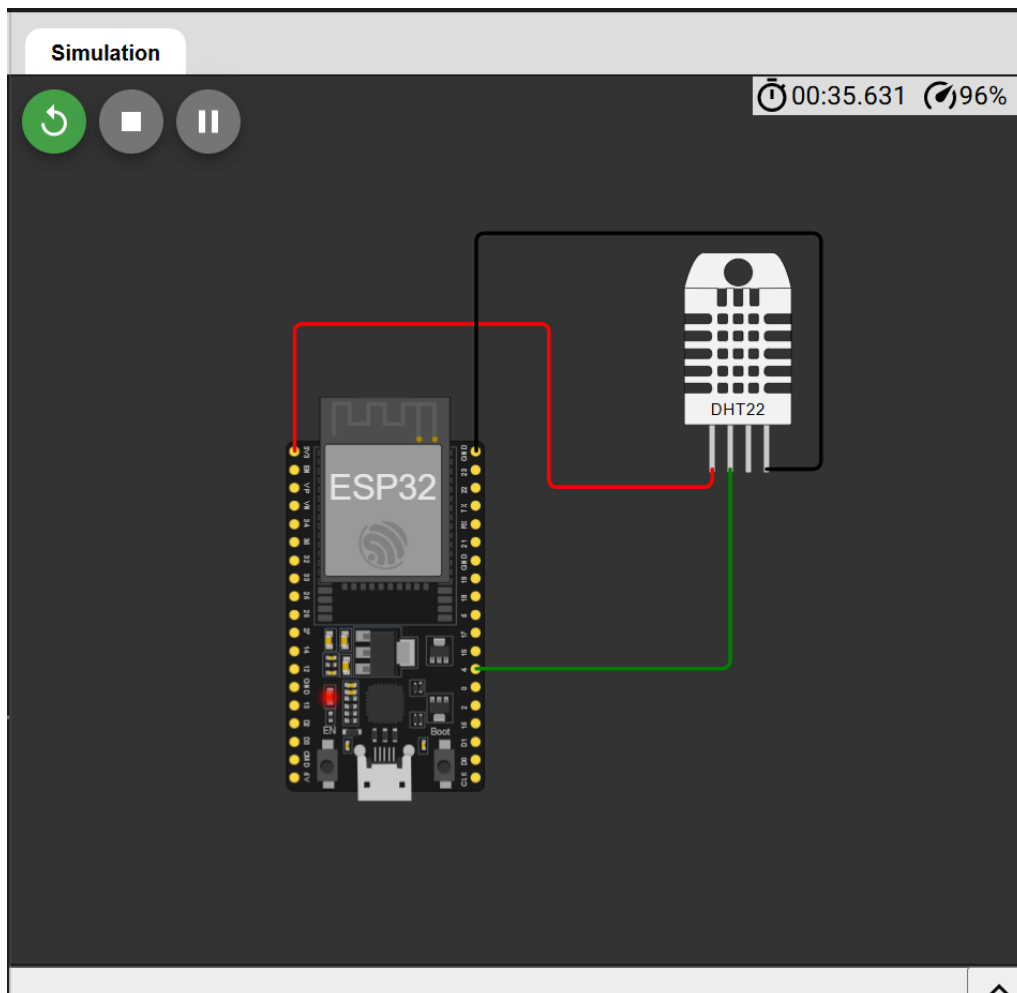
Serial.print(" Temperature : ");
Serial.print(temperature);
Serial.println(" °C");

Serial.print(" Humidity   : ");
Serial.print(humidity);
Serial.println(" %");

Serial.println("=====");
Serial.println();
}
}
}

```

Circuit Diagram and Output



```

-----
Smart Environmental Monitor
DecodeLabs - Project 1
-----
Reading Temperature & Humidity...

===== Sensor Readings =====
Temperature : 24.90 °C
Humidity    : 40.00 %
=====

===== Sensor Readings =====
Temperature : 24.90 °C
Humidity    : 40.00 %
=====

===== Sensor Readings =====
Temperature : 24.90 °C
Humidity    : 40.00 %
=====

```

Using I2C LCD:

```

#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <DHT.h>

```

```

#define DHTPIN 4
#define DHTTYPE DHT22

```

```

DHT dht(DHTPIN, DHTTYPE);
LiquidCrystal_I2C lcd(0x27, 16, 2);
unsigned long previousMillis = 0;
const unsigned long interval = 2000;

```

```

void setup() {
  Serial.begin(115200);

```

```

  dht.begin();

```

```

  lcd.init();
  lcd.backlight();

```

```

  lcd.setCursor(0, 0);
  lcd.print(" Smart Env Mon");
  lcd.setCursor(0, 1);
  lcd.print(" Initializing");

```

```

  Serial.println("=====");
  Serial.println(" Smart Environmental Monitor");
  Serial.println(" DecodeLabs Project 1");
  Serial.println("=====");

```

```

    delay(2000);
    lcd.clear();
}

void loop() {

    unsigned long currentMillis = millis();

    if (currentMillis - previousMillis >= interval) {

        previousMillis = currentMillis;

        float humidity = dht.readHumidity();
        float temperature = dht.readTemperature();

        if (isnan(temperature) || isnan(humidity)) {

            Serial.println("Failed to read from DHT sensor!");

            lcd.clear();
            lcd.setCursor(0, 0);
            lcd.print("Sensor Error");
            lcd.setCursor(0, 1);
            lcd.print("Check DHT22");

            return;
        }

        Serial.println("-----");
        Serial.print("Temperature : ");
        Serial.print(temperature);
        Serial.println(" C");

        Serial.print("Humidity   : ");
        Serial.print(humidity);
        Serial.println(" %");
        Serial.println("-----");

        lcd.clear();
    }
}

```

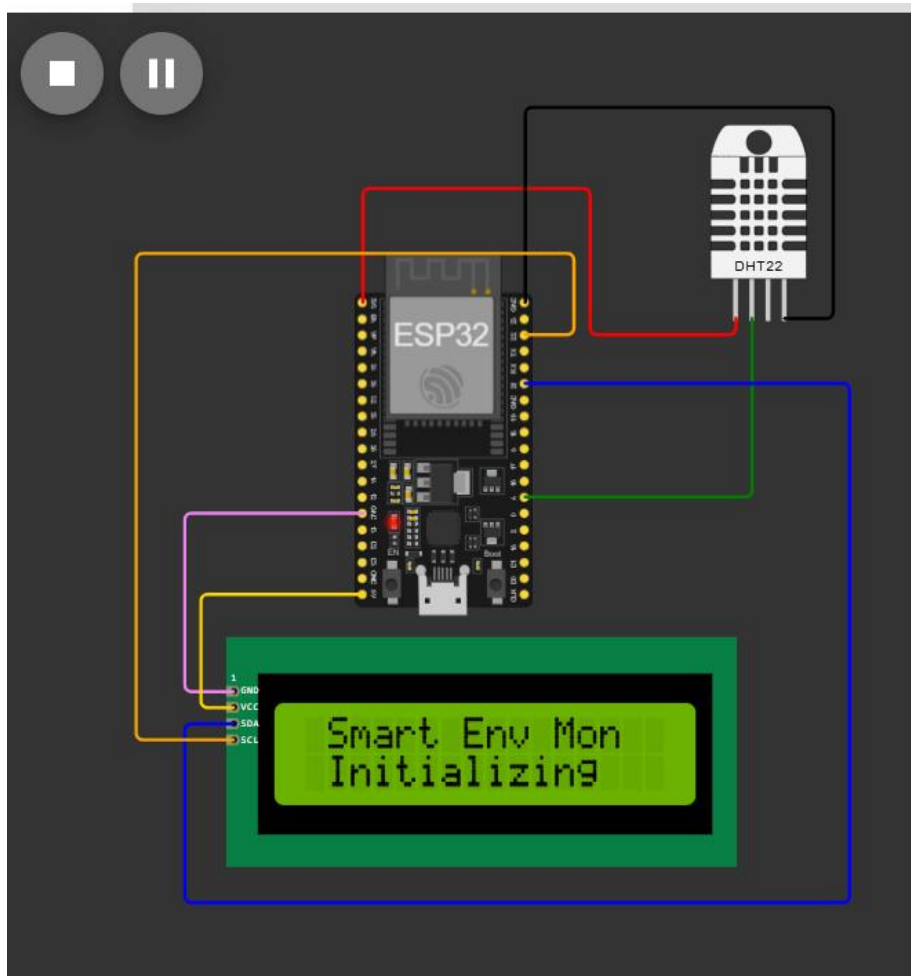
```

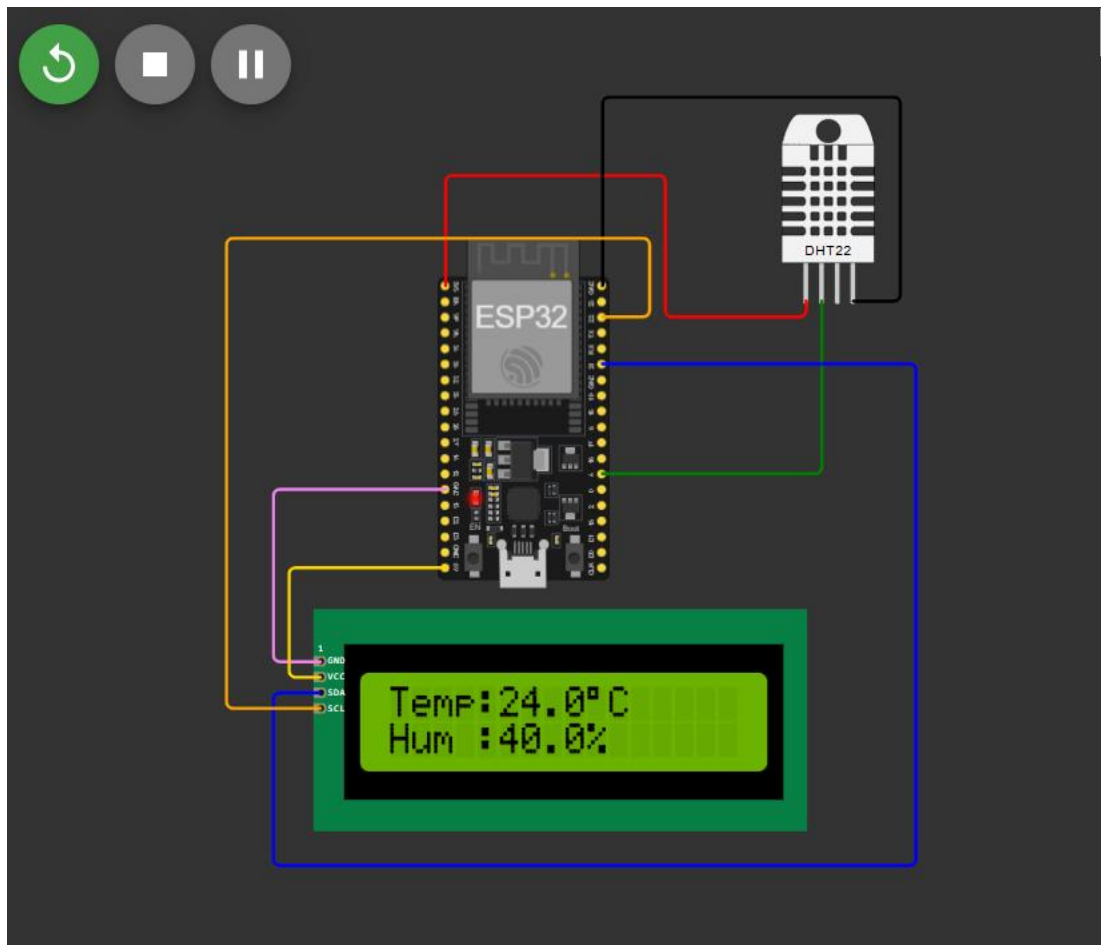
    lcd.setCursor(0, 0);
    lcd.print("Temp:");
    lcd.print(temperature, 1);
    lcd.print((char)223); // Degree Symbol
    lcd.print("C");

    lcd.setCursor(0, 1);
    lcd.print("Hum :");
    lcd.print(humidity, 1);
    lcd.print("%");
}
}

```

Circuit Diagram and Output





Conclusion

The Smart Environmental Monitor was successfully implemented using an ESP32 and DHT22 sensor. Temperature and humidity readings were acquired at regular intervals and displayed on both the Serial Monitor and the I2C LCD. The `millis()` implementation demonstrated a more efficient non-blocking approach than `delay()`, making it suitable for real-time IoT applications where multiple tasks must execute simultaneously.